

Lecture 02 : HTML Nested Lists & Tables

Nested Lists, Tables, colspan, rowspan & Complex Table Structures

Table of Contents

- ▶ 1.0 Overview
- ▶ 2.0 Nested Lists Concept
- ▶ 3.0 Nested Unordered List (Country)
- ▶ 4.0 Mixed Nested Lists (City)
- ▶ 5.0 Course List (Nested UL)
- ▶ 6.0 Daily Meal Plan (Complex Nested)
- ▶ 7.0 Tables in HTML
- ▶ 8.0 Basic Table Structure
- ▶ 9.0 Table Head, Body & Footer
- ▶ 10.0 colspan Attribute
- ▶ 11.0 rowspan Attribute
- ▶ 12.0 Complex Table (5-Level)
- ▶ 13.0 Homework Tasks
- ▶ 14.0 Quick Revision Checklist

1.0 Overview

Lecture 02 mein hum do powerful concepts cover karenge: **Nested Lists** (list ke andar list) aur **Tables** (rows, columns, colspan, rowspan). Ye concepts aapko professional-level web pages banane mein help karenge.

2.0 Nested Lists Concept

PATTERN

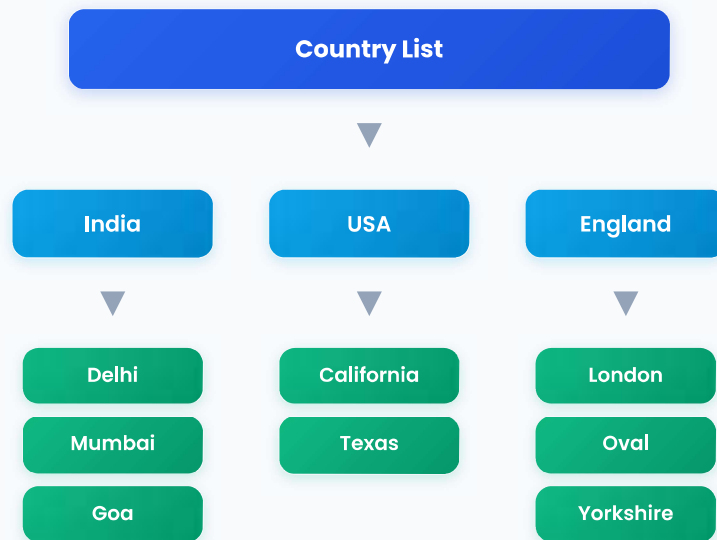
Nested List = List ke andar ek aur List

Jab aap kisi `<li>` ke andar ek aur `<ul>` ya `<ol>` daalte ho, toh wo ban jaata hai **Nested List**. Ye concept **for loop ke andar for loop** (nested loop) jaisa hai.

2.1 Nested List ka Matlab

Ab tak humne simple lists seekhi thi — ek level ki list. Lekin real-world mein data hierarchical hota hai:

Hierarchical Data Structure



INTUITION

Socho ek family tree jaisa — Grandfather ke andar Father, Father ke andar Son. Har level pe ek aur level nested hai. Same concept HTML lists mein apply hota hai.

3.0 Nested Unordered List — Country Example

3.1 Visual Output (Kaisa Dikhega)

Country List

- India
 - Delhi
 - Mumbai
 - Goa
- USA
 - California
 - Texas
- England
 - London
 - Oval
 - Yorkshire

3.2 Code Structure

```
HTML

<ul>  <!-- Outer List Start -->
  <li>India
    <ul>  <!-- Inner List Start -->
      <li>Delhi</li>
      <li>Mumbai</li>
      <li>Goa</li>
    </ul>  <!-- Inner List End -->
  </li>

  <li>USA
    <ul>
      <li>California</li>
      <li>Texas</li>
    </ul>
  </li>

  <li>England
    <ul>
      <li>London</li>
      <li>Oval</li>
      <li>Yorkshire</li>
    </ul>
  </li>
</ul>  <!-- Outer List End -->
```

IMPORTANT

Key Rule: Inner list hamesha `<li>` ke andar hi aati hai. Aap `<li>` ke andar `<ul>` ya `<ol>` daal sakte ho. `<li>` ke bahar direct `<ul>` nahi daalna.

4.0 Mixed Nested Lists — City Example

Ab hum dekhenge ki **Ordered List** ke andar **Unordered List** kaise daalte hain. Ye mixed nesting hoti hai.

4.1 Visual Output

Cities with Areas

- India
 - Delhi
 1. Dwarka
 2. Vikaspuri
 - Mumbai
 - Goa
- USA
 - California
 - Texas
 1. Austin
 2. Dallas
- England
 - London
 - Oval
 - Yorkshire

4.2 Code Structure

```
HTML

<ul>
  <li>India
    <ul>
      <li>Delhi
        <ol> <!-- Ordered List inside -->
          <li>Dwarka</li>
          <li>Vikaspuri</li>
        </ol>
      </li>
      <li>Mumbai</li>
      <li>Goa</li>
    </ul>
  </li>

  <li>USA
    <ul>
      <li>California</li>
      <li>Texas
        <ol>
          <li>Austin</li>
          <li>Dallas</li>
        </ol>
      </li>
    </ul>
  </li>
</ul>
```

TAKEAWAY

Mixed Nesting: Aap `<ul>` ke andar `<ol>` daal sakte ho, aur `<ol>` ke andar `<ul>` daal sakte ho. Koi restriction nahi hai. Bas dhyan rakho ki inner list hamesha `<li>` ke andar hi aani chahiye.

5.0 Course List — Nested UL Example

5.1 Visual Output

Gigs for Geeks — Course List

- DSA
 - Array
 - Linked List
 - Stack
 - Queue
- Web Technology
 - HTML
 - CSS
 - JavaScript
- Aptitude
- Gate
- Placement

5.2 Code Structure

```
HTML

<ul>
  <li>DSA
    <ul>
      <li>Array</li>
      <li>Linked List</li>
      <li>Stack</li>
      <li>Queue</li>
    </ul>
  </li>

  <li>Web Technology
    <ul>
      <li>HTML</li>
      <li>CSS</li>
      <li>JavaScript</li>
    </ul>
  </li>

  <li>Aptitude</li>
  <li>Gate</li>
  <li>Placement</li>
</ul>
```

6.0 Daily Meal Plan — Complex Nested List

Ye ek complex example hai jisme **Ordered List** ke andar **Headings (H2)** aur **Unordered Lists** dono hain.

6.1 Visual Output

Daily Meal Plan

1. **Breakfast**

- Main
 - Maple Syrup
 - Fresh Berries
 - Whipped Cream
- Side
- Drink

2. **Lunch**

- Main Course
 1. Starter
 2. Add
 3. Place
 4. Finish
- Side
- Beverages

3. **Dinner**

- Appetizer
- Main Course
 1. First
 2. Next
 3. Finally
- Dessert

6.2 Code Structure

```
HTML

<h1>Daily Meal Plan</h1>

<ol>
  <li><h2>Breakfast</h2>
    <ul>
      <li>Main
        <ul>
          <li>Maple Syrup</li>
          <li>Fresh Berries</li>
          <li>Whipped Cream</li>
        </ul>
      </li>
      <li>Side</li>
      <li>Drink</li>
    </ul>
  </li>

  <li><h2>Lunch</h2>
    <ul>
      <li>Main Course
        <ol>
          <li>Starter</li>
          <li>Add</li>
          <li>Place</li>
          <li>Finish</li>
        </ol>
      </li>
      <li>Side</li>
      <li>Beverages</li>
    </ul>
  </li>

  <li><h2>Dinner</h2>
    <ul>
      <li>Appetizer</li>
      <li>Main Course
        <ol>
          <li>First</li>
          <li>Next</li>
          <li>Finally</li>
        </ol>
      </li>
      <li>Dessert</li>
    </ul>
  </li>
</ol>
```

BEST PRACTICE

Semantic Structure: Breakfast, Lunch, Dinner ko H2 heading di gayi hai kyunki ye same level pe hain (H1 ke baad H2). Ye heading hierarchy follow karta hai.

7.0 Tables in HTML

PATTERN

Table = Rows + Columns

HTML mein table create karne ke liye hum `<table>` tag ka use karte hain. Table ke andar **rows** hoti hain (horizontal) aur **columns** hoti hain (vertical).

7.1 Table Terminology

Term	HTML Tag	Meaning
Table	<code><table></code>	Poora table container
Table Row	<code><tr></code>	Ek horizontal row
Table Data	<code><td></code>	Ek cell mein data
Table Header	<code><th></code>	Heading cell (bold + centered)
Table Head	<code><thead></code>	Header section of table
Table Body	<code><tbody></code>	Main content section
Table Footer	<code><tfoot></code>	Footer/conclusion section

Table Structure Hierarchy



8.0 Basic Table Structure

8.1 Visual Output

Name	Role
Rohit	Front End Developer
Mohit	Back End Developer
Anjali	Full Stack Developer

8.2 Code Structure

```
HTML

<table border="1">
  <!-- Row 1: Header -->
  <tr>
    <th>Name</th>
    <th>Role</th>
  </tr>

  <!-- Row 2: Data -->
  <tr>
    <td>Rohit</td>
    <td>Front End Developer</td>
  </tr>

  <!-- Row 3: Data -->
  <tr>
    <td>Mohit</td>
    <td>Back End Developer</td>
  </tr>

  <!-- Row 4: Data -->
  <tr>
    <td>Anjali</td>
    <td>Full Stack Developer</td>
  </tr>
</table>
```

IMPORTANT

Flow Samajhna: Sabse pehle `<table>` tag likhte hain. Uske andar `<tr>` (table row) likhte hain. Har `<tr>` ke andar `<td>` (table data) ya `<th>` (table header) likhte hain. `<th>` automatically bold aur centered hota hai.

9.0 Table Head, Body & Footer

Professional tables mein hum table ko teen parts mein divide karte hain: **Head** (headings), **Body** (actual data), aur **Footer** (conclusion/summary).

9.1 Visual Output

Item	Quantity	Price
Apple	10	\$1
Orange	5	\$0.5
Total	15	\$1.5

9.2 Code Structure

HTML

```
<table border="1">

  <!-- TABLE HEADER -->
  <thead>
    <tr>
      <th>Item</th>
      <th>Quantity</th>
      <th>Price</th>
    </tr>
  </thead>

  <!-- TABLE BODY -->
  <tbody>
    <tr>
      <td>Apple</td>
      <td>10</td>
      <td>$1</td>
    </tr>
    <tr>
      <td>Orange</td>
      <td>5</td>
      <td>$0.5</td>
    </tr>
  </tbody>

  <!-- TABLE FOOTER -->
  <tfoot>
    <tr>
      <td>Total</td>
      <td>15</td>
      <td>$1.5</td>
    </tr>
  </tfoot>

</table>
```

INTERVIEW TIP

Why use thead, tbody, tfoot? Ye sirf visual ke liye nahi hai. Screen readers (jo blind users use karte hain) in tags ko padhke samajh jaate hain ki "ye heading hai", "ye body data hai", "ye footer/summary hai". SEO mein bhi ye tags help karte hain.

10.0 colspan Attribute

PATTERN

colspan = Column Span (Ek cell ko multiple columns mein spread karna)

Jab aapko ek cell ko **do ya zyada columns** mein spread karna ho, toh `colspan="number"` attribute use karte hain.

10.1 Visual Output

Name	Contact Info	
John Doe	99-345-678-910	john@example.com
Cooper	12345678910	cooper@example.com
Mohan	10	mohan@gmail.com

10.2 Code Structure

HTML

```
<table border="1">
  <thead>
    <tr>
      <th>Name</th>
      <th colspan="2">Contact Info</th>
      <!-- This header spans 2 columns -->
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>John Doe</td>
      <td>99-345-678-910</td>
      <td>john@example.com</td>
    </tr>
    <tr>
      <td>Cooper</td>
      <td>12345678910</td>
      <td>cooper@example.com</td>
    </tr>
  </tbody>
</table>
```

WARNING

Common Mistake: colspan="2" likhne ke baad bhi agar aap 3 <td> daalte ho, toh table ka structure bigad jaayega. Jitna colspan hai, utne hi cells us row mein hone chahiye.

11.0 rowspan Attribute

PATTERN

rowspan = Row Span (Ek cell ko multiple rows mein spread karna)

Jab aapko ek cell ko **do ya zyada rows** mein spread karna ho, toh `rowspan="number"` attribute use karte hain.

11.1 Visual Output

Column 1	Column 2	Column 3
Row 1 Cell 1	Row 1 Cell 2	Row 1 Cell 3
	Row 2 Cell 2	Row 2 Cell 3
Row 3 Cell 1 (spans all 3 columns)		

11.2 Code Structure

HTML

```
<table border="1">
  <thead>
    <tr>
      <th>Column 1</th>
      <th>Column 2</th>
      <th>Column 3</th>
    </tr>
  </thead>

  <tbody>
    <!-- Row 1: Cell 1 spans 2 rows -->
    <tr>
      <td rowspan="2">Row 1 Cell 1</td>
      <td>Row 1 Cell 2</td>
      <td>Row 1 Cell 3</td>
    </tr>

    <!-- Row 2: No need for Column 1 cell -->
    <tr>
      <!-- Column 1 is already covered by rowspan -->
      <td>Row 2 Cell 2</td>
      <td>Row 2 Cell 3</td>
    </tr>

    <!-- Row 3: Cell 1 spans all 3 columns -->
    <tr>
      <td colspan="3">Row 3 Cell 1</td>
    </tr>
  </tbody>
</table>
```

IMPORTANT

rowspan Rule: Jis row mein rowspan="2" laga hai, uske neeche wali row mein us column ka cell **mat daalo**. Kyunki wo cell already upar wale ne cover kar liya hai. Isliye neeche wali row mein ek kam <td> hoga.

12.0 Complex Table — 5-Level Hierarchy

Ye sabse complex example hai jisme **colspan** aur **rowspan** dono ek saath use hote hain.

12.1 Visual Output

Level 1	Level 2	Level 3	Level 4	Level 5
System	System Apps	System ENV	App Test	Foo
			App Memory	Bar
				Bar
	System Memory			Foo
	System ENV 2	Memory Function		Foo
		FS Test		Bar

12.2 How to Build This Table — Step by Step

WORKFLOW

Step 1: Count Maximum Rows & Columns

Sabse pehle diagram ko dhyan se dekho aur count karo:

- **Maximum Rows:** 7 (header + 6 body rows)
- **Maximum Columns:** 5 (Level 1 se Level 5)

WORKFLOW

Step 2: Build Header Row

Header mein 5 columns hain — Level 1 se Level 5. Simple <th> tags se banao:

```
<thead>
  <tr>
    <th>Level 1</th>
    <th>Level 2</th>
    <th>Level 3</th>
    <th>Level 4</th>
    <th>Level 5</th>
  </tr>
</thead>
```

WORKFLOW

Step 3: Build Body Rows One by One

Har row ko carefully analyze karo:

```
<tbody>
  <!-- ROW 1: System (rowspan=6), System Apps (rowspan=4),
        System ENV (rowspan=3), App Test, Foo -->
  <tr>
    <td rowspan="6">System</td>
    <td rowspan="4">System Apps</td>
    <td rowspan="3">System ENV</td>
    <td>App Test</td>
    <td>Foo</td>
  </tr>

  <!-- ROW 2: App Memory (rowspan=2), Bar -->
  <tr>
    <td rowspan="2">App Memory</td>
    <td>Bar</td>
  </tr>

  <!-- ROW 3: Bar (App Memory already covered) -->
  <tr>
    <td>Bar</td>
  </tr>

  <!-- ROW 4: System Memory (colspan=2), Foo -->
  <tr>
    <td colspan="2">System Memory</td>
    <td>Foo</td>
  </tr>

  <!-- ROW 5: System ENV 2 (rowspan=2), Memory Function (colspan=2), Foo -->
  <tr>
    <td rowspan="2">System ENV 2</td>
    <td colspan="2">Memory Function</td>
    <td>Foo</td>
  </tr>

  <!-- ROW 6: FS Test (colspan=2), Bar -->
  <tr>
    <td colspan="2">FS Test</td>
    <td>Bar</td>
  </tr>
</tbody>
```

IMPORTANT

Complex Table Strategy:

1. Pehle maximum rows aur columns count karo
2. Har row ko left se right analyze karo
3. Jahan bhi cell merge ho raha hai, wahan colspan ya rowspan lagao
4. Jo cell upar wale ne cover kar liya hai, usko next row mein mat likho
5. Step-by-step build karo, ek row complete hone ke baad hi agli row pe jao

13.0 Homework Tasks

WORKFLOW

Task 1: Nested List Homework

Ek complex nested list banao jisme:

- Outer list ho
- Har item ke andar ek aur ya ho
- Minimum 3 levels ki nesting ho

WORKFLOW

Task 2: Complex Table Homework

Ek table banao jisme:

- thead, tbody, tfoot structure ho
- Minimum 2 colspan use ho
- Minimum 2 rowspan use ho
- Total minimum 5 columns aur 6 rows ho

14.0 Quick Revision Checklist

- Nested List = List ke andar ek aur list
- Inner list hamesha ke andar aati hai
- ke andar daal sakte hain (mixed nesting)
- ke andar daal sakte hain
- Nesting multiple levels deep ho sakti hai
- Table = <table> tag se start hota hai
- <tr> = Table Row (horizontal)
- <td> = Table Data (cell content)
- <th> = Table Header (bold + centered by default)
- <thead> = Header section (headings)
- <tbody> = Body section (actual data)
- <tfoot> = Footer section (summary/conclusion)
- colspan = ek cell ko multiple columns mein spread karta hai
- rowspan = ek cell ko multiple rows mein spread karta hai
- rowspan wale column ka cell next row mein mat daalo
- Complex table banane ke liye pehle max rows & columns count karo
- Step-by-step har row ko left se right build karo
- Screen readers thead/tbody/tfoot ko samajh sakte hain